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POSTURE CHECK: A COMPUTER VISION-BASED SYSTEM FOR REAL-TIME EXERCISE POSTURE MONITORING AND FEEDBACK

Computer Vision Project Proposal

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Problem Statement

Many people perform exercises with improper posture or form, which can reduce the effectiveness of their workouts and increase the risk of injury. This can be especially challenging for beginners, who may have difficulty determining whether they are performing an exercise correctly without the guidance of a trainer. Therefore, there is a need for an accessible system that can monitor exercise posture and provide immediate feedback. To meet this need, this project aims to develop a computer vision-based exercise posture detection system that uses a camera to analyze a user's body position while performing different exercises. The system will identify key body points, compare the user's posture with predefined correct positions, and provide feedback when improper form is detected. It will focus on common exercises such as squats, push-ups, lunges, and jumping jacks, helping users improve their form and perform exercises more safely and effectively.

Objectives

General Objective:

To develop a computer vision-based exercise posture detection system that can monitor a user's exercise form and provide feedback to help them perform exercises correctly and safely.

Specific Objectives:

1. To detect and track key body points using computer vision while the user performs exercises.
2. To analyze the user's posture and movement during exercises such as squats, push-ups, lunges, and jumping jacks.
3. To identify incorrect exercise postures by comparing the user's movements with predefined correct postures.
4. To provide real-time feedback to users when improper posture or form is detected.
5. To evaluate the system using manually verified test samples and report posture-classification accuracy, landmark-detection reliability, and processing speed in frames per second (FPS).

Expected Outcomes

The project is expected to produce a computer vision-based exercise posture detection system capable of monitoring a user's body movements during exercise. Specifically, the system is expected to:

1. Detect and track body key points from webcam frames with visibility checks for the landmarks used in posture evaluation.
2. Analyze selected exercise modes for squats, push-ups, lunges, and jumping jacks using exercise-specific posture rules.
3. Determine whether the user's posture is correct or incorrect based on predefined posture standards.
4. Provide real-time feedback when improper posture or form is detected.
5. Help users improve their exercise form by guiding them toward safer and more effective movements.
6. Target at least 80% correct/incorrect posture-classification accuracy on a manually verified held-out test set and at least 10 FPS on the development laptop under controlled, full-body webcam conditions. If the full four-exercise scope is not reliable, the minimum deliverable will retain a stable squat posture evaluator with real-time feedback and repetition counting.

Proposed Approach

The project will adopt a pose-estimation-based approach to detect and evaluate a user's body posture while performing exercises. The system will use Python, MediaPipe Pose Landmarker, and OpenCV to process webcam footage, extract body landmarks, analyze movements, and provide real-time posture feedback. Instead of training a custom deep learning model from scratch, the system will use a pretrained pose-estimation model and geometric, rule-based posture evaluation. The exercise mode will be selected from the supported exercises so the corresponding posture rules can be applied consistently.

- Baseline Method: MediaPipe Pose Landmarker with direct, single-frame joint-angle and alignment thresholds. This provides a simple working reference that can detect basic posture conditions without temporal processing.
- Main Method: MediaPipe landmarks with exercise-specific geometric rules, temporal smoothing, movement-phase tracking, corrective feedback, and repetition counting. This improves the baseline by reducing frame-to-frame instability and duplicate detections while producing more useful real-time feedback.

Algorithms

1. **Pose Estimation — MediaPipe Pose Landmarker**

MediaPipe Pose Landmarker will serve as the primary algorithm for detecting human body landmarks. It will identify key points, including the shoulders, elbows, wrists, hips, knees, and ankles, from each video frame. The coordinates and visibility information of these landmarks will provide the necessary data for analyzing body position and movement during an exercise.

2. **Video Processing and Visualization — OpenCV**

   OpenCV will handle webcam access, video frame acquisition, and the visualization of the detected body landmarks. It will also be used to display the pose skeleton, posture classification results, and corrective feedback on the processed video. It will serve as the connection between the webcam input, pose-estimation algorithm, and user interface.

3. **Joint Angles and Body Alignment — Geometric Feature Extraction**

   The system will calculate geometric features, including joint angles, relative distances, and body alignment. Joint angles will be calculated using the coordinates of three relevant landmarks. These measurements will serve as the primary features for evaluating exercise posture.

4. **Posture Evaluation — Rule-Based Classification**

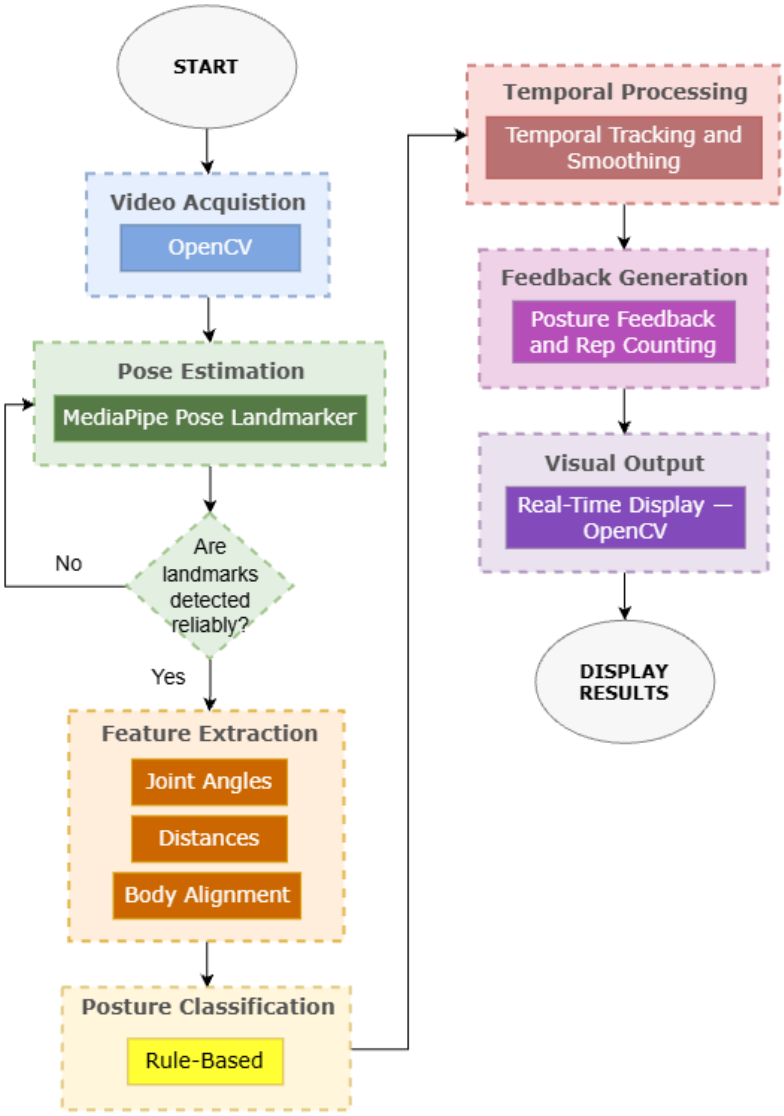
   The main classification method will be a rule-based algorithm that compares the extracted geometric features against predefined thresholds and exercise-specific conditions. Each supported exercise will have its own set of evaluation rules based on relevant joint angles, body alignment, and movement requirements. For instance, squat evaluation will consider knee and hip angles, while push-up evaluation will examine elbow movement and shoulder-hip-ankle alignment. If a measurement does not satisfy the applicable criteria, the system will identify the corresponding form issue and display an appropriate corrective message. The thresholds will be refined through testing and comparison with reference exercise guidelines.

5. **Temporal Tracking and Smoothing — Movement Consistency**

   The system will analyze landmark measurements and posture classifications across consecutive video frames to reduce inconsistent results caused by small fluctuations in landmark detection. Temporal tracking will also help identify exercise phases, such as the lowering and rising movements of a squat. By detecting completed movement cycles, the system can implement repetition counting and reduce duplicate counts caused by repeated detections within the same phase.

Proposed System Pipeline

The system will follow a sequential processing pipeline consisting of video acquisition, pose estimation, feature extraction, posture classification, temporal tracking, and feedback generation. First, OpenCV will capture frames from the webcam. MediaPipe Pose Landmarker will then detect the user's body landmarks, which will be used to calculate joint angles and other geometric features. The rule-based classifier will evaluate these features against the predefined criteria for the detected exercise. Temporal tracking will help stabilize the results and identify completed repetitions. Finally, OpenCV will display the processed video, pose skeleton, posture status, repetition count, and corrective feedback.



Workflow Diagram

Data

The attached dataset evidence contains 20 extracted image frames listed in labels.csv, with four exercise classes represented equally: 5 squat, 5 push-up, 5 lunge, and 5 jumping-jack frames. The frames come from four public Wikimedia Commons exercise videos. For this project, these samples will be used as feasibility/reference data and as an initial set for checking the pose-estimation and posture-evaluation pipeline; live webcam recordings will also be used during implementation and testing.

Data Sources and Licenses:

- Squat — 5 frames; source/author: FitnessScape; license: CC BY 3.0; link: https://commons.wikimedia.org/wiki/File:Squat_-_exercise_demonstration_video.webm
- Push-up — 5 frames; source/author: Centers for Disease Control and Prevention; license: Public domain (US federal government work); link: https://commons.wikimedia.org/wiki/File:Muscle_Strengthening_at_Home_-_Push-ups.webm
- Lunge — 5 frames; source/author: Skhulile Mthiyane; license: CC BY-SA 4.0; link: https://commons.wikimedia.org/wiki/File:Forward_lunge_training.webm
- Jumping jack — 5 frames; source/author: Taco fleur; license: CC BY-SA 4.0; link: https://commons.wikimedia.org/wiki/File:Jumping_jacks_and_burpees.webm

Labels and Annotation Status: The CSV already provides the exercise_label for all 20 frames and records the source video, extraction time, author, license, and label basis. However, form_correctness is marked UNVERIFIED for all 20 samples. Therefore, correct/incorrect posture labels will not be assumed. Before those samples are used to evaluate posture classification, they will be manually reviewed against the project's predefined exercise criteria and the verification decision will be recorded.

Coverage and Known Conditions: The evidence set covers four different source videos and four exercise categories. The push-up samples include wall and knee variants, while the other listed frames do not specify an exercise variant. The CSV does not explicitly encode lighting, camera angle, or background categories, so these conditions will be documented during manual review and supplemented with controlled webcam recordings that keep the full body visible.

Data Split: Because the project uses a pretrained pose model and rule-based thresholds rather than training a new classifier from scratch, the initial 20-frame set will be split in a stratified 60/20/20 manner for rule development, validation, and held-out testing: 12 development/reference frames (3 per exercise), 4 validation frames (1 per exercise), and 4 test

frames (1 per exercise). The final evaluation should use additional manually verified webcam clips so that performance is not judged from only four held-out still frames.

Timeline

The project will be developed according to the course schedule, with activities organized around the required checkpoints. Each phase will focus on completing specific project tasks, from initial planning and system design to implementation, testing, and final presentation.

Week	Activities and Tasks	Target Deliverable
Week 1: October 15	Finalize proposal sections, measurable objectives, baseline/main method, workflow, data evidence, timeline, and risks.	Proposal and Initial Plan
Week 2: October 17–22	Set up Python, OpenCV, and MediaPipe; test webcam and pose estimation; verify samples; start the feasibility spike and initial posture logic.	50% Progress Check
Week 3: October 23–29	Implement joint-angle features, baseline per-frame rules, and exercise-specific posture classification; begin controlled testing.	Core Implementation and Testing
Weeks 4–5: October 30–November 12	Add temporal smoothing, corrective feedback, and repetition counting; verify labels, test conditions, and measure accuracy and FPS.	80% Progress Check
Week 6: November 13–19	Finalize and test the system; prepare documentation and presentation materials; rehearse the demo and technical defense.	Final Presentation and Live Demo

Risk

The project may encounter technical challenges during implementation and testing. The two most likely risks and their corresponding fallback plans are identified below.

Risk 1: Inaccurate Body Landmark Detection

MediaPipe Pose Landmarker may fail to detect certain body landmarks accurately because of poor lighting, camera positioning, partial body visibility, or rapid movements. Inaccurate landmark detection may lead to incorrect joint-angle calculations and unreliable posture classifications.

- Fallback Plan: Adjust the camera position, lighting, and distance to ensure that the user's body remains visible within the frame. Implement landmark visibility checks and temporal smoothing to reduce unreliable predictions. If accurate detection remains difficult, limit the evaluation to exercises and body landmarks that can be reliably captured under the available setup.

Risk 2: Inaccurate Posture Classification

The rule-based classification algorithm may incorrectly identify a posture as correct or incorrect if the predefined angle thresholds and alignment rules do not adequately account for differences in body proportions, exercise technique, or camera perspective.

- Fallback Plan: Test and refine the posture thresholds using sample recordings and reference exercise guidelines. Prioritize the most measurable posture criteria for each exercise and focus on implementing reliable squat detection first. If accurate classification across all four exercises cannot be achieved within the project timeline, retain the core functionality for the exercises that perform reliably and document the limitations of the remaining exercises.